Lecture 10 · Part II · Extended notes

Explaining and auditing models

Dataset: Adult census income, 48,842 people

Why these notes exist

This lecture sits outside Géron, the course's primary text. For it *these notes are the primary source*, and it is examined on exactly the same terms as the textbook lectures. Everything in the slide deck is here, with the arguments written out at length rather than compressed onto a slide; the numbers are the ones the accompanying notebook prints, and every one of them is a measurement on the dataset named above rather than a figure quoted from a paper.

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1 Where we are

Every lecture so far asked how good a model is. Lecture 3 made the number honest — held-out rows, precision and recall, a threshold chosen on purpose; Lecture 4 asked whether a probability of 0.7 is right seven times in ten; Lecture 5 compared trees, forests and boosting and measured how stable they are. A model that passes all of that can still rely on something nobody would defend, and can still fail one group of people while scoring well on everyone.

This lecture asks the two questions the test score cannot answer. *What does the model rely on?* — for the whole model, and for one prediction. *For whom does it fail?* — by slice of the population, by three fairness criteria, and through the proxies that carry an attribute the model was never given.

What you should be able to do afterwards.

- Say why impurity importance misleads, and compute and read permutation importance on held-out rows.
- Define partial dependence and ICE curves, and say how a partial dependence can misrepresent the model.
- Define the Shapley value of one prediction against a background, prove its efficiency property, and compute it exactly for a small number of features.
- Audit a classifier by slice; define demographic parity, equal opportunity and calibration within groups, and prove that they cannot all hold when base rates differ.
- Show that dropping a sensitive column does not remove its information, and say what a threshold per group buys and costs, and who decides it.

2 The problem

A bank's premium card needs an income above a band. Applicants answer a dozen questions in the bank's app, and a model estimates the probability that their income is above the band. At 0.5 or above the applicant is pre-approved on the spot; below, they are sent to a manual check of their payslips, which is slower and which many abandon. A qualified applicant sent to the slow path costs the applicant, and the bank a customer; an unqualified applicant pre-approved costs the bank when the documents arrive.

The product team removed `sex` and `race` from the model's inputs, reasoning that a model which never sees an attribute cannot use it. The lecture tests that reasoning.

2.1 The data

The Adult census income data, the 1994 US census extract on which most fairness methods have been tried, downloaded from OpenML by the notebook: one row per person, the target whether they earned more than 50K a year. The bank is a frame; the method transfers, the numbers do not.

Quantity	**Value**
People	48,842
Answers the model sees (of them categorical)	12 (6)
Fit / validation / test rows	29,304 / 9,769 / 9,769
Share of women	0.3315
Share above 50K: everyone, men, women	0.2393, 0.3038, 0.1093

The twelve answers are age, work class, education twice (as a level and as a number of years, `education-num`), marital status, occupation, the relationship in the household, capital gains and losses, hours worked and country — and `fnlwgt`, the census's sampling weight, a number about how the survey was drawn rather than about the person. A categorical answer becomes one integer code per level, and a missing answer stays missing: both models below handle a missing value natively.

The sensitive attributes are kept aside, not thrown away

`sex` and `race` are not given to the model. They stay beside every row, aligned with it, for the audit only; the split is stratified on the target and on sex together, so that every group the audit reads has the same share in the fit, validation and test rows. Sex is recorded by the census in two categories, and the audit inherits that.

2.2 The model

A histogram-based gradient-boosted ensemble, `HistGradientBoostingClassifier`, with the six categorical answers declared as categories so that a code is never read as a size. On the test rows its accuracy at 0.5 is 0.8714, against 0.7607 for refusing everyone; its AUC is 0.9248; it pre-approves 0.1988 of the applicants. It is a good model, and everything that follows is about what it does.

3 What the model relies on

3.1 Impurity importance, and why it misleads

The boosted model has no `feature_importances_`. A random forest of 200 trees fitted to the same rows has one: for each column, the impurity decrease of every split on it, weighted by the share of rows that reach the split, summed over the trees. It is measured on the training rows the trees were grown on, and that is its first defect.

Beside it, **permutation importance**, defined in the next subsection, computed twice: on the training rows and on held-out rows. The three agree about most columns and disagree completely about one. `fnlwgt` is first by impurity, 0.1793; sixth by permutation on the training rows, 0.0313; and eleventh on held-out rows, where shuffling it changes the AUC by −0.0006 — nothing.

Failure condition — importance earned on the training rows

Two causes, both measured. *The training rows:* a fully grown forest fits its own rows exactly (AUC 1.0000 there, 0.9046 on test rows), so every split, even one on noise, reduces impurity there, and permutation on those rows still credits what the trees memorised. *Many distinct values:* a column with a value per row offers a split anywhere, and deep in a tree some split on it always cleans a small node; `fnlwgt` takes 20,077 distinct values on the fit rows, age 73. Impurity importance measures how much a tree used a column to fit its training rows, not how much the predictions on new rows depend on it; and it exists only for trees.

3.2 Permutation importance

Definition 3.1 (Permutation importance)**.** *For a fitted model f, held-out rows (X, y) and a score such as the AUC, the permutation importance of column j is*

$$I_j = \mathrm{AUC}(f; X, y) - \frac{1}{R}\sum_{r=1}^{R} \mathrm{AUC}\big(f; X^{(\pi_r, j)}, y\big),$$

where $X^{(\pi_r, j)}$ is X with column j replaced by a random permutation π_r of itself.

Shuffling keeps the column's marginal distribution and destroys its link with the target and with the rest of the row. The measure needs nothing from the model but its predictions, so it works for any model; it costs R prediction passes per column; and the R repeats give a spread, which is reported beside the mean. On the boosted model, with ten repeats, capital gain, marital status and age come first and `fnlwgt` last.

4 The shape of an effect

Importance says how much the model relies on a column; it does not say how the prediction moves when the column changes. To see that for age, take one applicant, keep every answer they gave, change only their age — 20, 22, . . . , 80 — and ask the model each time. The predictions, joined, are that applicant's curve. Do it for every applicant and average the curves at each age: one curve, what the model predicts on average if everyone had that age.

Definition 4.1 (ICE and partial dependence)**.** *For applicant i and column j, the* ***individual conditional expectation*** *is the model's output with column j set to v and every other answer kept,* $\mathrm{ICE}_i(v) = f\left(v, \mathbf{x}_{-j}^{(i)}\right)$. *The* ***partial dependence*** *is its average over the applicants,* $\mathrm{PD}(v) = \frac{1}{n}\sum_{i=1}^{n} \mathrm{ICE}_i(v)$.

The notebook computes both by hand, in probabilities — the units of the product rule — on 1,000 test applicants and ages from 18 to 80. `sklearn.inspection.partial_dependence` does the same; for a boosted model its fast path averages the log-odds instead, which is a different curve.

4.1 A toy example

Before the data, a model written by hand, so that every number can be checked on paper: an applicant who is not married has probability 0.10 at any age; a married one has 0.10 at age 20, rising by 0.10 every ten years. Four applicants, two married and two not, each at their own age:

Applicant	Age	Married	Own prediction	ICE at 20	at 40	at 60
A	30	yes	0.20	0.10	0.30	0.50
B	55	yes	0.45	0.10	0.30	0.50
C	35	no	0.10	0.10	0.10	0.10
D	60	no	0.10	0.10	0.10	0.10
Partial dependence (mean of the column)				0.10	0.20	0.30

A's curve is read by setting A's age to 20, 40 and 60, keeping "married", and asking the model; A's real age only marks where A sits on that curve. From 20 to 60 the married curves rise by 0.40, the others by 0.00, and the partial dependence by 0.20 — a rise no applicant has. The average is half the married curve because half the applicants are married. The same happens on the census data, below.

4.2 What a partial dependence hides

Age	Partial dependence	Married	Everyone else
20	0.015	0.030	0.003
30	0.203	0.385	0.055
50	0.270	0.490	0.092
70	0.218	0.367	0.097

The partial dependence is exactly the weighted average of the two groups' mean curves, with the share married, 0.447, as the weight. The married go from 0.030 at 20 to 0.490 at 50; everyone else from 0.003 to 0.092. The effect of age depends on another answer, and the average of two such curves describes nobody.

Failure condition — an average over an interaction

Where the effect of one column depends on another, the partial dependence averages curves of different shapes, and can average a rise and a fall into a flat line. Draw the ICE curves, coloured by the suspected interaction.

5 One prediction: the Shapley value

Examinable: the definition, efficiency and its proof, the linear check.

5.1 The question

Of the 826 test applicants who earn above 50K and were refused, take the one at the median probability — a typical refusal, chosen by a rule rather than by eye. He is 50, married, a high-school graduate; his occupation and work class were left blank; he has no capital gains or losses and works 40 hours. The model gives him 0.2880, under the threshold. Under the GDPR he may ask what in his answers counted against him. The answer we want is one number per answer that says how much it moved his probability, and the numbers should add up to the probability itself.

5.2 Definition

Treat the M features as players, $N = \{1, \dots, M\}$. A **coalition** $S \subseteq N$ is a set of features whose values are the applicant's; the others are filled in from a **background** sample B of applicants, and the model is averaged over it.

Definition 5.1 (Value function and Shapley value)**.**

$$v(S) = \frac{1}{|B|} \sum_{\mathbf{b} \in B} f\left(\mathbf{x}_S, \mathbf{b}_{\bar{S}}\right), \qquad \phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(M - |S| - 1)!}{M!} \left(v(S \cup \{i\}) - v(S)\right).$$

Here $(\mathbf{x}_S, \mathbf{b}_{\bar{S}})$ *is the row with the applicant's values on* S *and the background row's elsewhere.* $v(N) = f(\mathbf{x})$ *is the prediction, and* $v(\varnothing)$*, the background's mean prediction, is the* ***baseline****.*

5.3 The weights are an average over orders

Reveal the features one at a time, in an order π, a permutation of N. Let P_i^π be the set of features revealed before i. When i is revealed it adds $v(P_i^\pi \cup \{i\}) - v(P_i^\pi)$.

Proposition 5.2. $\phi_i = \dfrac{1}{M!}\displaystyle\sum_\pi \big(v(P_i^\pi \cup \{i\}) - v(P_i^\pi)\big)$, *the marginal contribution of i averaged over all $M!$ orders.*

Proof

Group the orders by the set that precedes i. For a given $S \subseteq N \setminus \{i\}$, the orders with $P_i^\pi = S$ are those that list the $|S|$ features of S in any order, then i, then the remaining $M - |S| - 1$ features in any order: there are $|S|!\,(M - |S| - 1)!$ of them, and each contributes the same term $v(S \cup \{i\}) - v(S)$. Dividing by $M!$ gives exactly the weight in the definition. □

5.4 Efficiency

Proposition 5.3 (Efficiency). $\displaystyle\sum_{i=1}^{M} \phi_i = v(N) - v(\varnothing) = f(\mathbf{x}) - \frac{1}{|B|}\sum_{\mathbf{b}\in B} f(\mathbf{b}).$

Proof, in three steps

1. Fix one order. Write $\pi = (i_1, i_2, \ldots, i_M)$ and $S_k = \{i_1, \ldots, i_k\}$, so $S_0 = \varnothing$ and $S_M = N$. The feature i_k is revealed after exactly S_{k-1}, so its contribution in this order is $v(S_k) - v(S_{k-1})$.

2. The sum telescopes. Over the M features, in this one order,

$$\sum_{k=1}^{M} \big(v(S_k) - v(S_{k-1})\big) = v(S_M) - v(S_0) = v(N) - v(\varnothing).$$

The value does not depend on π.

3. Average over the orders. By the proposition above,

$$\sum_i \phi_i = \frac{1}{M!}\sum_\pi \sum_i \big(v(P_i^\pi \cup \{i\}) - v(P_i^\pi)\big) = \frac{1}{M!}\sum_\pi \big(v(N) - v(\varnothing)\big) = v(N) - v(\varnothing).$$

The order of the two sums may be swapped because both are finite. Finally, $v(N) = f(\mathbf{x})$ because with every feature the applicant's, every background row gives the same input; and $v(\varnothing)$ is the background's mean prediction by definition. □

5.5 A check with a known answer

For a linear function $f(\mathbf{z}) = \mathbf{w}^\top \mathbf{z}$, $v(S) = \sum_{j\in S} w_j x_j + \sum_{j\notin S} w_j \bar{b}_j$, with $\bar{b}_j$ the background mean of column j. The marginal contribution of i is then $w_i(x_i - \bar{b}_i)$ whatever S is, and the weights over $S \subseteq N \setminus \{i\}$ sum to one — there are $\binom{M-1}{s}$ sets of size s, each of weight $s!\,(M-1-s)!/M!$, so each size contributes $1/M$ and the M sizes contribute 1. Hence $\phi_i = w_i(x_i - \bar{b}_i)$. The notebook asserts this before it explains the model.

5.6 The other properties

Beyond the syllabus, for context. The Shapley value also satisfies *symmetry* (two features that add the same to every coalition get the same value), *dummy* (a feature that adds nothing to any coalition gets zero) and *additivity* (the values for $f + g$ are those for f plus those for g). Shapley proved in 1953 that it is the only attribution with these four properties; that is the reason it is the attribution practitioners reach for.

5.7 Computed exactly

With $M = 12$ there are $2^{12} = 4{,}096$ coalitions. Against a background of 200 rows drawn from the fit set, one applicant needs $4{,}096 \times 200 = 819{,}200$ predictions, made in a single call; the value of every coalition is then a mean over its 200 rows, and each ϕ_i a weighted sum over the half of the coalitions that leave i out. The cost doubles with every feature, so exact enumeration stops being possible somewhere in the tens of features; sampling estimators, and exact algorithms specific to trees, are what the `shap` library implements.

5.8 His explanation

From the baseline 0.2568 to his 0.2880, the twelve values add up to 0.0312, exactly the difference:

Answer	Value	Reading
married	+0.0834	the largest push up
age 50	+0.0676	near the top of the age curve
occupation left blank	−0.0564	the largest thing held against him
HS-grad	−0.0428	below the background's education
no capital gain	−0.0334	most have none; the few with some score high
`fnlwgt`	−0.0052	a survey weight, moving one person's score

A blank occupation: 0.0559 of the fit rows have one, and 0.0947 of them earn above 50K against 0.2478 of the rest. The model has learned that a blank predicts a lower income. Whether a blank *should* cost an applicant anything is the bank's decision — and no global summary would have put that question on the table.

5.9 Local values, averaged

Averaged in absolute value over 40 test applicants, against the same background, the Shapley values give a global ranking that agrees with the permutation ranking at the top — marital status 0.0823, age 0.0550, capital gain 0.0498: averaged, many local answers give back a global one.

5.10 What each explanation cannot do

Method	Fails when	Seen here
Impurity importance	read on training rows; many distinct values	`fnlwgt` first, worth nothing
Partial dependence	an interaction	married against the rest
Shapley values	the background is a choice; filled-in rows can be impossible; the cost doubles per feature	the baseline is the background's mean, 0.2568
All of them	read as causes	they describe the function, not the person

Every tool in this section describes the model: what its output does when an input changes. None of them says what would happen to a person's income if their answer changed.

6 For whom does it fail

6.1 Metrics per slice

The first instrument of an audit is the familiar one, read per slice of the population rather than once over everyone, and always beside its count.

Test rows, threshold 0.5	n	Above 50K	Base rate	Accuracy	TPR	FPR
men	6530	1984	0.3038	0.8398	0.6613	0.0823
women	3239	354	0.1093	0.9352	0.5650	0.0194
age under 25	1678	13	0.0077	0.9940	0.3846	0.0012
age 25–44	4962	1225	0.2469	0.8587	0.6163	0.0618
age 45–64	2684	1003	0.3737	0.8141	0.6899	0.1118
age 65 and over	445	97	0.2180	0.8966	0.6186	0.0259

Accuracy points the wrong way. The model is more accurate for women, 0.9352 against 0.8398, and a qualified woman is pre-approved less often, 0.5650 against 0.6613: with a base rate of 0.1093, refusing is usually right, and accuracy rewards it. Under 25 the accuracy is 0.9940 and the TPR 0.3846 is five people out of 13. By race, the TPR is 0.6527 for the 2125 White applicants above 50K and 0.5702 for the 121 Black ones; the group recorded as Other has 6. A slice is read by a rate that conditions on the truth — TPR, FPR — and beside its count.

6.2 Three fairness criteria

Definition 6.1 (Three fairness criteria)**.** *With $\hat{Y}$ the decision, Y the truth, A the group and s the model's probability, for every group a:*

- ***demographic parity**: $P(\hat{Y} = 1 \mid A = a)$ is the same — the same share selected;*
- ***equal opportunity**: $P(\hat{Y} = 1 \mid Y = 1, A = a)$ is the same — the same TPR, so a qualified applicant's chance does not depend on the group;*
- ***calibration within groups**: $P(Y = 1 \mid s, A = a) = s$ — a probability means the same in every group. At a threshold its footprint is **predictive parity**, the same PPV, where PPV is Lecture 3's*

precision: of those selected, the share with $Y = 1$.

Measured on the shipped model at 0.5, the three give three verdicts. The share pre-approved is 0.2582 for men and 0.0790 for women, a ratio of 0.3061: demographic parity fails. The TPR is 0.6613 against 0.5650, a gap of 0.0963: equal opportunity fails. The PPV is 0.7782 against 0.7812: predictive parity holds. The FPR is 0.0823 against 0.0194. And the model is calibrated within both groups: the men's mean probability is 0.3019 against a base rate of 0.3038, the women's 0.1165 against 0.1093, with calibration errors (the count-weighted mean gap between a bin's mean probability and its observed share) of 0.0146 and 0.0137. A pre-approval means the same thing for a man and for a woman; a qualified woman gets one less often, because qualified women score lower — the model sees fewer of the answers that go with high incomes among them.

6.3 They cannot all hold

Examinable.

Proposition 6.2. *For any decision rule and any group with base rate $p = P(Y = 1 \mid A = a) > 0$,*

$$TPR \cdot p = selected \cdot PPV.$$

Consequently, if two groups have different base rates, demographic parity, equal opportunity and predictive parity cannot all hold.

Proof

1. Count the true positives twice. In a group of n people, the true positives are the selected people with $Y = 1$. As a share of those with $Y = 1$ they are $\text{TP} = \text{TPR} \cdot p\,n$; as a share of those selected they are $\text{TP} = \text{PPV} \cdot \text{selected} \cdot n$. Equating and dividing by n gives the identity.

2. Hold the three equal. Suppose the selection rate is q, the PPV is c and the TPR is t in both groups, with $c, t > 0$. Then $t\,p_1 = q\,c = t\,p_2$, so $p_1 = p_2$. Contrapositive: different base rates rule out all three at once.

3. Any two fix the third. With equal PPV and equal TPR, the identity forces $q_1/q_2 = p_1/p_2$; with equal selection and equal PPV it forces $t_1/t_2 = p_2/p_1$; with equal selection and equal TPR it forces $c_1/c_2 = p_1/p_2$. □

On the census data the ratio of base rates, women to men, is 0.3598. A rule that kept PPV and TPR equal would have to select women at 0.3598 times the men's rate; the shipped model, with PPV equal and TPR not, selects them at 0.3061 times. No model changes a base rate, and only equal base rates escape.

Beyond the syllabus, for context. Stronger versions hold at the level of scores. Kleinberg, Mullainathan and Raghavan (2016) show that calibration within groups and equal average scores among the positives and among the negatives of each group cannot hold together unless the base rates are equal or the predictor is perfect; Chouldechova (2017) derives the companion identity $\text{FPR} = \frac{p}{1-p} \cdot \frac{1-\text{PPV}}{\text{PPV}} \cdot \text{TPR}$, which forbids equal PPV, equal TPR and equal FPR together when base rates differ.

What the result buys you

No model satisfies every criterion. Choosing which one to hold is a decision about whom to protect, and it belongs to whoever is accountable for the product, with the numbers in front of them. The data team's job is to measure all of them.

6.4 Proxies

The model was never given sex. Fit the same model class with sex as the *target*, from the twelve answers it does see:

Predicting sex from	Test AUC
the twelve answers	0.9327
without `relationship`	0.8682
without `relationship` and `marital-status`	0.8030

`relationship` holds the levels Husband and Wife, which name the sex of every married applicant; hours and occupation carry much of the rest. No single column carries it, and removing columns lowers the AUC without taking it to 0.5.

Failure condition — fairness through unawareness

A model that never sees an attribute can still use it through its proxies. A bank that never records the attribute cannot even measure whether its model does. Keep it, under separate access, for the audit.

6.5 A threshold per group

The men's threshold stays at 0.5; the women's is chosen on the validation rows and the test rows are read once. For *equal opportunity*, the highest women's threshold at which their validation TPR reaches the men's: 0.3846. For *demographic parity*, the highest at which their validation selection rate reaches the men's: 0.0783.

Test rows	One threshold	Equal opportunity	Demographic parity
Women's threshold	0.5000	0.3846	0.0783
Women pre-approved	0.0790	0.0985	0.2658
TPR, women (men 0.6613)	0.5650	0.6525	0.9040
PPV, women (men 0.7782)	0.7812	0.7241	0.3717
FPR, women (men 0.0823)	0.0194	0.0305	0.1875
Accuracy, everyone	0.8714	0.8713	0.8341

Equal opportunity pre-approves 31 more qualified women and 32 more who do not qualify, at almost no cost in accuracy; demographic parity, 120 more qualified and 485 more who do not, and the PPV of a woman's pre-approval falls to 0.3717.

A threshold per group is a policy decision. It reads the attribute at the moment of decision — the model never saw sex, and the rule now reads it for every applicant — and anti-discrimination law

may forbid exactly that in credit. Every option has a price, which the table states. And there are remedies upstream that need no attribute at decision time: better answers on the form, the blank occupation treated differently, a manual review for everyone in a band below the threshold. The data team measures every criterion per group, with counts, shows what each remedy costs and buys, and writes it down; the choice is made, and recorded with its reasons, by the people accountable for it.

7 In practice

7.1 What to look out for

Situation	What goes wrong	Do this
The library's importance chart	impurity, on training rows: an identifier or a weight comes first	permutation importance on held-out rows, with its spread
One partial dependence curve	it averages curves of different shapes	ICE curves coloured by the suspect
A Shapley value quoted alone	its background was never stated	state the background and the baseline; check that the values add up
An explanation in a letter	it cites a proxy, or something nobody can change	review the reasons before they are sent
Accuracy per group	the group with the lower base rate looks best served	TPR and FPR per group, beside the base rate
A small slice	a rate on a dozen people swings by chance	print the count; pool, or wait, before acting
"We don't use sex"	proxies carry it; nobody can check	keep the attribute under separate access, for the audit
One criterion, passed	the others fail unseen	report all three; say which is held, and why
Thresholds tuned on the test rows	the equality was chosen, not measured	choose on validation, read the test rows once
An audit done once	the population drifts; a retrained model is a new model	re-audit on every retrain; monitor slice metrics

7.2 A model card

A model card is the one-page record that travels with a model: what it is for and not for, the data, the performance overall and by group, the known limits, the explanations available, and who owns its decisions. For the pre-approval model: intended for instant pre-approval, where a refusal sends the applicant to a manual check and never to a final no; not for credit limits, pricing or any decision without a manual path; fitted on 29,304 rows of census-style answers with sex and race excluded from the inputs and kept for the audit; test AUC 0.9248, accuracy 0.8714 against 0.7607 for refusing everyone; TPR 0.6613 for men and 0.5650 for women, PPV 0.7782 and 0.7812, calibrated in both groups; proxies for sex at an AUC of 0.9327, a blank occupation lowers the score, slices with under a hundred positives not assessed; permutation importance on held-out rows and Shapley values against a stated background of 200 fit rows; and a named owner for the fairness criterion held and the threshold rule, with the date and the reasons.

8 The notebook

The notebook runs on the CPU, top to bottom, and downloads the census data itself. It fits the bank's model; the forest and its three importances; permutation importance by hand; ICE and partial dependence by hand, on a toy model and on the data; Shapley values by full enumeration, checked on a linear function and by efficiency, for one applicant and for forty; and the audit — slices, the three criteria and the identity, calibration per group, proxies, and three threshold policies. The numbers in these notes are its run.

What to change

In the Shapley section, draw the background from the refused applicants instead of the fit rows. The baseline drops, every value changes, and they still add up to the same probability minus the new baseline — the background is part of the explanation.

9 Design exercise: auditing a credit-limit model before it is filed

The lecture closes with a design exercise in the manner of an ML system-design interview, on the product this lecture's methods are built for.

The task. A digital bank operating across the European Union sells a credit card inside its app. The feature is the model that decides, at application, whether to offer the card and with what credit limit, and later whether to raise it. The users are applicants and cardholders; the business measures losses from defaults, the share of applicants who take the card, and the revenue a limit earns. Under the EU AI Act a creditworthiness model is a high-risk system: before it ships, the bank must document it, examine its data for bias, and put human oversight and monitoring in place. Given: application answers, the bank's account history, credit-bureau data, and repayment outcomes for the cards it issued. The consumers of the output are the app, the credit officers who review cases, and the letters that tell an applicant why; the regulators are the financial supervisor, the data protection authority and anti-discrimination law.

1. **Frame the audit: decisions, harms, groups.** Name the decisions — offer or refuse, the limit, each raise — and the harm each error does: a good customer refused or kept at a low limit, a customer given more credit than they can repay. Name the groups the audit will read before any number is computed, and write the metrics per harm. *Pitfall:* auditing the overall score, which can be good on everyone and hide the group the model fails.

2. **The attributes the audit needs.** The model takes no protected attribute; the audit needs them anyway. Collect them with a legal basis and the applicant's knowledge, store them apart from the features with access limited and logged, and measure the proxies by predicting each attribute from the features. *Pitfall:* "we do not collect it, so we cannot discriminate" — postcode, employer, household answers and spending carry it, and without the attribute nobody can show whether the model uses it.

3. **Who was never labelled.** Repayment is observed only for applicants the old policy approved, and the refused are not a random sample. Keep a small randomised approval slice within risk limits, and say plainly which groups the outcomes cover thinly. *Pitfall:* auditing only the

approved, so that a group the old policy refused often has its error rates resting on a handful of people, or on none.

4. **What the model relies on.** Permutation importance on held-out applicants, with its spread, and a column of random numbers as a control. *Pitfall:* filing the library's impurity chart, on which an identifier, a timestamp or a sampling weight comes first and is read by the supervisor as what drives the decision.

5. **Shapes, and shortcuts.** ICE curves for every important feature, coloured by the suspected interaction; a search for features that encode how the data were collected rather than whether the customer repays; every important feature reviewed with the credit officers. *Pitfall:* one partial dependence curve filed alone, which averages customers whose curves have different shapes and describes none of them.

6. **Reasons for one applicant.** Shapley values for every refusal and every reduced limit, against a stated and versioned background; the largest negative values become plain-language reasons from a reviewed list; the values are checked to add up on every letter. *Pitfall:* a reason that is a proxy or that nobody can act on — a proxy is a finding for the audit, not a reason.

7. **The audit by group.** Per group, the offer rate, the TPR among customers who repay, the FPR among those who default, calibration, and the limits granted, each with its count and an interval; all three criteria reported with the identity. *Pitfall:* reporting the one criterion that passes, when a calibrated model fails equal opportunity whenever the base rates differ.

8. **Remedies, oversight, monitoring.** Remedies upstream first; any rule that reads a protected attribute at decision time decided by the legal and policy owners and recorded; a model card, logs of every decision and its reasons, credit officers able to overrule with the override logged, slice metrics monitored, and a re-audit on every retrain. *Pitfall:* a per-group threshold chosen by the data team, which may be unlawful and which nobody accountable approved.

9. **File it.** Decisions, harms and groups named; attributes kept apart for the audit and proxies measured; the unlabelled refused accounted for; permutation importance with groups and a control; ICE curves and a shortcut review; reviewed reasons from Shapley values; every criterion per group, with counts; remedies decided by their owners; model card, logs, oversight and monitoring — each step named for the pitfall it avoids.

10 Exercises

The five set at the end of this lecture's deck, in the style of the written paper. The solutions are on Lecture 11's deck and in the exercise book.

1. A random forest ranks a census sampling weight first by impurity importance, yet shuffling that column on held-out rows does not change the AUC. Give the two reasons impurity importance can rank such a column first. [4]

2. The partial dependence of a model's probability on age rises from 0.015 at 20 to 0.270 at 50. Say what the curve averages, and how it can misrepresent the model. [4]

3. Define the Shapley value ϕ_i of feature i for one prediction, given a value function v over coalitions of the M features, and prove that $\sum_i \phi_i = v(N) - v(\varnothing)$. [5]

4. A classifier is applied to two groups whose base rates differ, and at its threshold the two groups have the same PPV and the same selection rate. Show that they cannot also have the same true-positive rate. [5]

5. A bank removes the applicant's sex from its model's inputs and reports that the model is therefore fair. State two things wrong with the claim, and what the bank should do instead. [4]

Summary

Two questions a test score cannot answer: what a model relies on, and for whom it fails. Impurity importance ranked the census's sampling weight first, at 0.1793, earned on the training rows; permutation importance on held-out rows put it at nothing. A partial dependence averages ICE curves of different shapes and can describe nobody. The Shapley value of one prediction, computed exactly over 4,096 coalitions, adds up to the prediction minus the baseline — that is efficiency, proved by a telescoping sum over orders — and found that a blank on the form was the largest thing held against one applicant. The audit read every rate per slice with its count: accuracy pointed the wrong way, the three fairness criteria gave three verdicts, and an identity on the true positives shows they cannot all hold when base rates differ. Dropping sex left proxies that predict it at an AUC of 0.9327; a threshold per group buys equal opportunity at a price the table states, and is a decision for the people accountable for the product. The closing design exercise audited a digital bank's credit-limit model before it is filed as a high-risk system, in nine steps, each named for the pitfall it avoids.